

AI-Driven Personal Finance Assistant

Product (working name): FinPilot AI

Stack: MERN + OpenAI

Timeline: 3 weeks (production-deployable MVP)

1) PRODUCT REQUIREMENTS DOCUMENT (PRD)

1.1 Vision

Deliver an AI co-pilot that:

- Tracks income & expenses
- Auto-categorizes transactions
- Forecasts cash flow
- Provides personalized financial advice

1.2 Target Users

- Salaried professionals (22–40)
- Freelancers
- Early-stage entrepreneurs

1.3 Success Criteria (MVP)

- Users can onboard and add transactions in < 3 minutes
- AI gives contextual advice in < 4 seconds
- 30% of users use AI chat within first session
- 60% 7-day retention (beta)

2) MARKET & UX BENCHMARKS

Use these strictly as reference:

Platform	What to Learn
Mint	Auto categorization + clean dashboard
YNAB	Budget discipline UX
Personal Capital	Net worth visualization
Cleo	Conversational AI tone

Platform	What to Learn
Revolut	Modern fintech UI system

We combine:

- Mint's tracking
- YNAB's structure
- Cleo's AI chat
- Revolut's UI polish

3) MVP SCOPE (STRICT — 3 WEEKS)

Included

- ✓ Authentication
- ✓ Manual transactions
- ✓ AI categorization +Auto-categorizes transactions

- ✓ AI chat assistant
- ✓ Dashboard analytics
- ✓ Monthly forecast
- ✓ Subscription system
- ✓ Investment portfolio
- ✓ Credit score

4) SOFTWARE ARCHITECTURE (SADD)

4.1 Tech Stack

Frontend

- React (Vite)
- TailwindCSS
- ShadCN
- Recharts
- React Query

Backend

- Node.js

- Express.js
- MongoDB Atlas
- JWT Auth
- Zod validation
- Stripe

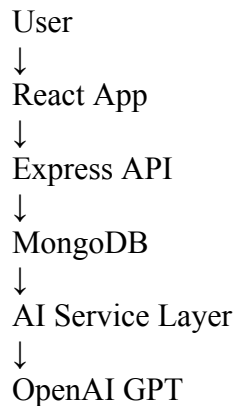
AI Layer

- OpenAI GPT API
- Structured prompt templates

Deployment

- Frontend: Vercel
- Backend: Render
- DB: MongoDB Atlas

4.2 Architecture Flow



5) DATABASE DESIGN (FINALIZED)

User Schema

- name
- email (unique)
- passwordHash
- subscriptionTier (free/pro)
- monthlyIncome
- createdAt

Indexes:

- email unique index
- userId reference index

Transaction Schema

- userId (indexed)
- amount
- category
- merchant
- type (income/expense)
- date
- notes
- isRecurring

Goal Schema

- userId
- title
- targetAmount
- currentAmount
- deadline

6) FUNCTIONAL REQUIREMENTS

6.1 Authentication

- Register/login
- JWT access + refresh
- Password reset
- Secure HTTP-only cookies

6.2 Transaction Module

User can:

- Add/edit/delete transaction
- Filter by category/date
- Import CSV (basic)

AI auto-categorizes merchant name.

6.3 Dashboard

Must show:

- Total income (current month)
- Total expense
- Savings %
- Category pie chart
- Line chart (income vs expense)
- AI financial score

6.4 AI Intelligence

A) Auto Categorization

Prompt:

“Classify this merchant into a financial category: Starbucks”

Return JSON:

```
{  
category: "Dining"  
}
```

B) Cash Flow Forecast

Formula:

Current balance

- predicted income
– predicted expenses

C) AI Chat

User asks:

- “Can I afford a MacBook?”
- “Why is my savings low?”
- “How to improve my financial health?”

Context injected:

- Monthly income
- Expense breakdown
- Savings %
- Goals

Guardrails:

- No guaranteed investment advice
- No illegal financial schemes
- Add financial disclaimer

7) NON-FUNCTIONAL REQUIREMENTS

- API response < 400ms (non-AI)
- AI response < 4s
- 99% uptime
- 70% test coverage
- Secure password hashing
- Rate limiting
- GDPR delete/export

8) UX / UI SPECIFICATION

Design Principles

- Minimal fintech
- Soft shadows
- Strong data hierarchy
- Mobile-first

Color System

Primary: Emerald

Accent: Indigo

Background: Light gray

Screen Breakdown

1 Onboarding

- Enter income
- Set savings goal
- Add first transaction

2 Dashboard

- Net balance card
- Income vs expense graph
- Category pie chart
- AI suggestion card

3 Transactions Page

- Table view
- Filter by date
- Edit inline
- Bulk delete

4 Goals Page

- Progress bar
- Deadline tracker
- AI optimize button

5 AI Assistant

- Chat interface
- Quick prompts
- Conversation history

9) SUBSCRIPTION MODEL

Free

- 10 transactions/month
- 5 AI queries/month

Pro (\$12/month)

- Unlimited transactions
- Unlimited AI
- Forecasting
- Advanced analytics

Stripe webhook handles subscription status update.

10) SECURITY ARCHITECTURE

- HTTPS only
- JWT with rotation
- Zod validation
- Input sanitization
- MongoDB schema validation
- Rate limiting middleware

- CORS policy strict

11) TESTING STRATEGY

Unit:

- Auth
- Transactions
- Forecast calculation

Integration:

- AI endpoint
- Stripe webhook

E2E:

- Signup → Add transaction → Chat → Upgrade plan

12) DEV ROADMAP (3 WEEKS STRICT)

Week 1 – Core Foundation

Backend

- Auth
- User schema
- Transactions CRUD
- MongoDB setup

Frontend

- Auth pages
- Dashboard layout
- Transaction table

Deliverable:

✓ Users can register and add transactions.

Week 2 – Intelligence Layer

Backend

- AI service wrapper
- Categorization endpoint
- Forecast logic
- Financial score formula

Frontend

- AI chat UI
- Charts integration
- Goals page

Deliverable:

✓ AI categorizes & chats with context.

Week 3 – Monetization & Production

Backend

- Stripe integration
- Subscription middleware
- Rate limits

Frontend

- Subscription page
- Upgrade modal
- Error handling

Final

- Deployment
- Performance testing
- Security audit

Deliverable:

✓ Production-ready SaaS.